

TUTORIAL – 5: Defect Reporting

Aim

To identify software defects and prepare a professional **Bug (Defect) Report** including **Bug ID, Severity, Priority, Environment, Screenshots, and Steps to Reproduce** using a sample application.

Description

A **software defect (bug)** is an error or flaw in a software application that causes it to produce incorrect or unexpected results or behave differently from its specified requirements.

Defect Reporting is the process of documenting software defects so that developers can understand, reproduce, fix, and verify them.

A good bug report should contain:

- Bug ID
- Bug Summary
- Module Name
- Severity
- Priority
- Environment
- Steps to Reproduce
- Expected Result
- Actual Result
- Screenshot
- Status

Proper defect reporting improves communication between testers and developers and helps deliver high-quality software.

Problem Statement

Test the **Login Module** of the **Student Management System** and identify any software defects. Prepare a professional Bug Report with severity, priority, reproduction steps, and supporting evidence (screenshot).

How to Perform This Practical (Student Procedure)

Step 1: Open the Application

Run the Student Management System in your IDE or browser.

Step 2: Read the Requirements

Requirement:

- Username should not be empty.
- Password should not be empty.
- Login should succeed only with valid credentials.

- Invalid login should display an error message.

Step 3: Execute Test Cases

Try different inputs.

Test Case	Input
Valid Login	admin / admin123
Invalid Password	admin / 12345
Blank Username	""
Blank Password	""
Invalid Username	student / admin123

Step 4: Identify Defects

Example Defects

- ✓ Login accepts blank username
- ✓ Error message not displayed
- ✓ Login button not responding
- ✓ Password visible instead of hidden

Step 5: Record Bug Details

Prepare a bug report including

- Bug ID
- Summary
- Severity
- Priority
- Environment
- Screenshot
- Steps to Reproduce

Step 6: Attach Screenshot

Take a screenshot whenever a defect occurs.

Example:
Login Page

Username :

Password :

[Login]

(No error message displayed)

Screenshot Attached

Step 7: Submit Bug Report

Share the report with the developer for fixing.

Implementation Details

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
Version	Version 1.0
Programming Language	Python
IDE/Tool Used	Visual Studio Code / PyCharm
Database Used	MySQL
Operating System	Windows 10
Browser (If Applicable)	Google Chrome

Test Environment	Testing Environment
Tester Name	Student Name
Test Date	DD/MM/YYYY

Source Code

```
username = input("Enter Username : ")
password = input("Enter Password : ")
```

```
# Defect: Blank username/password validation is missing
if username == "admin" and password == "admin123":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

Bug Identification Details

Field	Details
Bug ID	BUG-001
Defect Title	Login accepts blank username
Module Name	Login Module
Feature Affected	User Authentication
Defect Description	The application allows the Login button to be clicked even when the Username field is blank. Proper validation is missing.
Defect Type	Functional Defect
Severity	Medium
Priority	High
Environment	Windows 10, Google Chrome
Reported By	Student Name

Date Reported	DD/MM/YYYY
----------------------	------------

Steps to Reproduce

Step No.	Action Performed	Input/Test Data	Expected Result	Actual Result
1	Open Login Page	-	Login page displayed	Login page displayed
2	Leave Username blank	""	Validation message should appear	No validation
3	Enter Password	admin123	Username required	Password accepted
4	Click Login	Click	Login should not proceed	Error shown only after submission

Bug Report Template

Field	Details
Bug ID	BUG-001
Bug Summary	Blank username validation missing
Module Name	Login Module
Severity	Medium
Priority	High
Steps to Reproduce	Leave username blank and click Login
Expected Result	"Username Required" message should appear
Actual Result	No validation message appears before login attempt

Assigned To	Developer
Status (New/Open/In Progress/Fixed/Closed)	Open
Resolution	Pending
Retest Status	Not Yet Tested
Remarks	Validation needs to be implemented

Bug Summary

Bug ID	Module	Severity	Priority	Status
BUG-001	Login	Medium	High	Open
BUG-002	Login	High	High	Open
BUG-003	Registration	Medium	Medium	Fixed

Observation

Sr. No.	Observation	Remarks
1	Login works with valid credentials	Pass
2	Blank username validation missing	Defect Found
3	Invalid password displays correct message	Pass
4	Password field should mask entered characters	Suggested Improvement
5	Bug report prepared successfully	Completed

Output

Testing Completed

Bug Report Generated

Bug ID : BUG-001
Module : Login
Severity : Medium
Priority : High
Status : Open

Screenshot Attached

Bug Submitted Successfully

Screenshot Placeholder:

Students should capture and attach a screenshot of the defect (e.g., missing validation message or incorrect behavior) in the lab manual after the Output section.

Solve the Below Mention Questionnaires

1. What is a software defect?

- A) A new software feature
- B) A problem that causes the software to behave incorrectly**
- C) A software update
- D) A testing tool

Answer: B) A problem that causes the software to behave incorrectly

2. Which document is used to record and track software defects?

- A) User Manual
- B) Test Plan
- C) Bug (Defect) Report**
- D) Project Schedule

Answer: C) Bug (Defect) Report

3. Severity in a bug report indicates:

- A) The order in which the defect should be fixed
- B) The impact of the defect on the system**

- C) The name of the tester
- D) The number of test cases executed

Answer: B) The impact of the defect on the system

4. Which section of a bug report helps developers reproduce the defect?

- A) Module Name
- B) Steps to Reproduce**
- C) Resolution
- D) Priority

Answer: B) Steps to Reproduce

5. Why are screenshots included in a bug report?

- A) To increase the size of the report
- B) To provide visual evidence of the defect**
- C) To improve software performance
- D) To reduce the number of test cases

Answer: B) To provide visual evidence of the defect

Conclusion / Outcome

- Understood the concept of software defects and defect life cycle.
- Identified defects in the Login Module through testing.
- Prepared a professional Bug Report with Bug ID, Severity, Priority, Environment, and Reproduction Steps.
- Learned the importance of screenshots and clear documentation in helping developers reproduce and resolve defects.
- Gained practical experience in defect reporting using standard software testing practices.